Winter term 2020/2021: Exercises "Radio Frequency Technology"

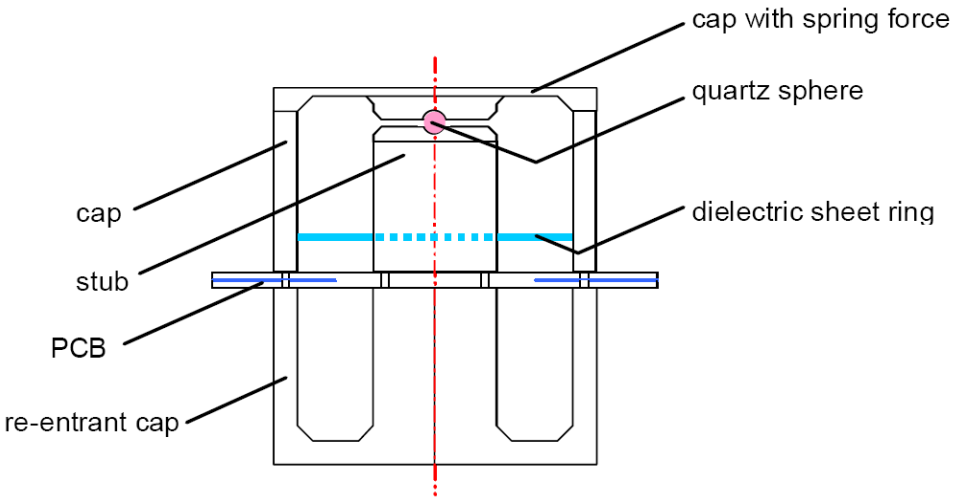
EXERCISE SESSION 10 (22.01.2021)

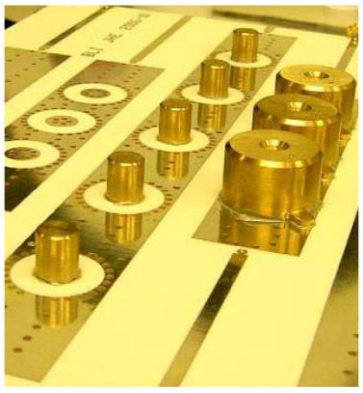
Problem 10.1: Coaxial resonator

A re-entrant coaxial resonator in SMT-compatible technology consists of several parts which contribute to resonator loss and therefore decrease Q_0 in different amounts. For a given structure, simulation allows to "switch off" all loss contributions but one and therefore to quantify these.

a) From the data given below, calculate the overall Q_0.
b) Assuming gold metallization (σ = 41 x 10^6 S/m) of the caps instead of silver (σ = 61.7 x 10^6 S/m), calculate the resulting Q_0.

part metalization	Ag	$Q_0 = 3210$
PCB traces / vias	Cu	$Q_0 = 7740$
PCB dielectric loss	$\tan\delta = 0.0015$	$Q_0 = 6610$
sphere dielectric loss	$\tan\delta = 0.0002$	$Q_0 \approx 100000$
dielectric layer loss	$\tan\delta = 0.001$	$Q_0 = 6215$





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Problem 10.2: Cylindrical cavity feed and Q factor

A microstrip line shall be coupled to a cylindrical TM_{010} cavity resonator to realize bandpass behavior at the resonance frequency.

(a) For a frequency of 6 GHz, an unloaded Q_o=10000 and silver conductivity of the cavity $\sigma = 61{,}7 \cdot 10^6 \frac{\mathrm{S}}{\mathrm{m}}$, calculate its dimensions.
(b) The microstrip lines shall couple through slots in their ground plane to this resonator. Draw locations, shape and orientation of the coupling structure with respect to the microstrip lines and to the cavity.

Problem 10.3: Cylindrical cavity feed and Q factor

The TE_{011} resonance results in particularly high Q factors.

(a) For a frequency of 6 GHz, determine the shape of a TE_{011} cavity with maximum Q and calculate this Q ($\sigma = 61{,}7 \cdot 10^6 \frac{\mathrm{S}}{\mathrm{m}}$).
(b) The resonator shows a second resonance at the same frequency. Name this mode. What can be done to avoid the resonator resonating at this unwanted mode?

Short questions session 10

a) Which type of cavity is generally best for higher Q factors: rectangular cavity or cylindrical cavity?

b) Which cylindrical waveguide mode gives the best Q factors when resonators are made from it?

c) Where would you place a slot feed for feeding a cylindrical waveguide in the fundamental mode?